

Gradient-Sensitive Functional Descriptors for Interferometric Surface Metrology: Complementing Amplitude-Averaging ISO Parameters

Abstract

This paper presents a complete interferometric pipeline for optical evaluation of roundness and surface profiles using a self-built Michelson interferometer. The method is validated on two certified JENOPTIK standards: the roundness standard FN 111 (24,000 interferograms) and the magnification standard FN 101 with a certified Flick flat (100,000 interferograms), each measured in five 400° sweeps. External validation against 104 tactile reference measurements yields an E_n score of 0.61 for the FN 111 and 0.3% agreement for the FN 101, demonstrating metrological consistency of the optical reconstruction. In addition to conventional ~~ISO-profile roughness parameters (ISO 21920-2, formerly ISO 4287 roughness parameters)~~ and roundness metrics ($RONt$, $RONq$), the reconstructed nanometre profiles are analysed using functional-analytic descriptors including L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) norms. Controlled defect-injection experiments and analysis of the certified Flick flat show that gradient-based descriptors provide complementary sensitivity to localised geometric features that may remain below the detection threshold of amplitude-averaging parameters. For a 100 nm synthetic spike, BV density increased by 18% and the $W^{1,2}$ seminorm by 622%, while R_a showed no detectable response at the reconstruction resolution. The BV descriptor also detected the certified Flick flat in a sliding-window analysis where R_a remained insensitive. These results suggest that functional descriptors can complement standard profile and roundness parameters, providing additional information on localised geometric variations without requiring changes to the measurement hardware. [All reported](#)

uncertainties are within-run repeatability estimates from five sweeps of a single continuous rotation and should be interpreted as lower bounds; an indicative type-B uncertainty budget is provided.

Keywords: interferometry, roundness, surface profile, functional norms, bounded variation, Sobolev seminorm, uncertainty

1. Introduction

Surface topography is one of the most important properties of technical parts. The accuracy of form and roughness influences contact conditions, wear and lifetime. Roundness, understood as deviation from a perfect circle, remains a central requirement in precision manufacturing; Zhu et al. emphasised that precise measurements are needed both for form tolerances and for tribological surface texture assessment [1]. At nanometric scale, such measurements require high-precision optical systems [2]. The foundational reference for surface metrology terminology and parameter definitions is the handbook of Whitehouse [3].

Optical dimensional metrology has a long connection with interference [4]. Modern optical surface metrology relies on diffraction, polarisation and interference [5, 6]. Diffraction and scattering have been applied to roughness measurement ~~[7–11], while interference in several distinct ways:~~ Brodmann introduced a scattered-light roughness instrument for production environments [7] and subsequently extended the scattered-light principle to combined roughness, form and waviness measurement [8]; Fan and Huynh analysed light scattering from flat periodic rough surfaces to estimate roughness parameters [9]; Hertzsch et al. demonstrated model-based scatterometry for microtopographic analysis of turned surfaces [10]; and Kucharski and Zdunek reported a low-cost scatterometry setup achieving nanometric precision in fast roughness measurement [11]. Interference remains central for ultra-precise surface-form measurement ~~[12, 13] and,~~ from the early interference-based length comparisons of Benoît [12] to modern large-scale precision optics programmes [13]. De Groot systematised the principles of phase-shifting interferometry is widely used in and coherence-scanning interference microscopy

25 for topography evaluation [14–16][14, 16], and Chatterjee described phase-evaluation
26 interferometric testing of reflective surfaces [15].

27 Several interferometric instruments have been proposed for high-accuracy
28 surface and roundness measurement. Polack et al. described a Michelson-based
29 system for surface-shape determination built around a high-resolution stitching
30 interferometer developed by Thomasset et al. [17, 18]. Fan et al. presented
31 a roundness measuring system for microspheres based on two small Michel-
32 son interferometers [19]. Kuhnel et al. proposed an interferometric system for
33 roundness and cylindricity measurement of ring gauges with 0.1 nm resolution
34 over a large measurement range [20]. Commercial instruments can be highly ac-
35 curate, but they are often costly and may not expose the full acquisition chain
36 needed for experimental method development.

37 The ~~present work~~ numerical analysis of fringe patterns has its own methodological
38 lineage, which frames the phase-decoding route used here. Takeda et al. introduced
39 the Fourier-transform method of fringe-pattern analysis, in which a spatial
40 carrier allows the phase to be recovered from a single interferogram without
41 temporal phase shifting [21]. Kemaio extended this concept with the two-dimensional
42 windowed Fourier transform, improving robustness to local fringe-quality variations
43 at the cost of window-parameter selection [22]. More recently, Feng et al. applied
44 deep learning to single-frame fringe analysis, training convolutional networks to
45 estimate the intermediate quantities of the arctangent phase calculation and
46 reporting accuracy approaching multi-frame phase shifting [23]. Galaktionov
47 and Toporovsky proposed a multi-stage fringe-map reconstruction algorithm
48 for fibre-end surface analysis in non-phase-shifting interferometry, showing that
49 geometric fringe processing remains competitive where phase-shifting hardware
50 is unavailable [24]. The radial excess-fraction method used in the present work
51 belongs to this single-frame, non-phase-shifting family: each interferogram is
52 decoded independently through the geometry of its circular fringes, which suits
53 continuous-rotation acquisition where temporal phase stepping is impractical.

54 The present work uses a self-built, non-contact Michelson-type roundness
55 measurement system assembled from off-the-shelf components. The optical

56 setup, apparatus description and radial image-processing route for phase extrac-
 57 tion were published in a previous study [25], where only roughness parameters
 58 were reported for the ceramic standard FN 111. The present work extends that
 59 study in three directions: (i) roundness ($RONt$, $RONq$) is evaluated for the first
 60 time with this self-built optical setup, (ii) a second artefact — the JENOPTIK
 61 magnification standard FN 101 with a Flick flat — is measured and validated
 62 against its DAkkS-DKD certificate, and (iii) a controlled sensitivity-comparison
 63 methodology is introduced: synthetic defects of known geometry are injected
 64 into measured profiles, and the response of classical ISO parameters is com-
 65 pared against that of L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) norms
 66 — standard tools of functional analysis here applied to interferometric profile
 67 metrology — all computed from the same nanometre residual. The comparison
 68 framework, rather than any individual norm, constitutes the methodological
 69 contribution.

70 The computational problem is that modern interferometric surface metrol-
 71 ogy records long image sequences before a numerical profile exists. In the present
 72 data, however, each PNG file is already a separate interferogram in which the
 73 phase is encoded spatially by the bright and dark fringes inside the image. The
 74 files in the FN 111 and FN 101 folders are therefore not treated as a temporal
 75 phase-shifting stack of one fixed scene. Instead, each image $I_i(x, y)$ is processed
 76 independently to obtain one wrapped excess-fraction value ϵ_i . The ordered se-
 77 quence of these values is then unwrapped over image order and converted to a
 78 nanometre profile. The full measurement archive contains 24,000 grayscale in-
 79 terferograms of the FN 111, with image size 648×488 px, and 100,000 grayscale
 80 interferograms of the JENOPTIK magnification standard FN 101, with image
 81 size 640×480 px.

82 The literature relevant to this comparison is mature but only partly con-
 83 nected. ~~Interferometric testing has developed approaches for stitching, registration~~
 84 ~~, motion-error compensation, and systematic-error control [26–30]. Repeatability~~
 85 ~~is commonly evaluated~~ In interferometric testing, Dumas et al. extended the
 86 lateral range of interferometry through subaperture stitching [26]; Chen et

87 al. developed an iterative stitching algorithm for spherical surfaces [27] and
 88 subsequently applied stitching to the testing of large optical surfaces [28]; Iacchetta
 89 et al. addressed rotation and translation registration of band-limited interferometric
 90 images using a chirp z-transform [29]; and Burge et al. applied computer-generated
 91 holograms to the measurement of x-ray and EUV optics [30]. On repeatability,
 92 Quan et al. compared evaluation approaches for interferometric surface repeatability [31]
 93 , and Zhang et al. compared repeatability across different interference systems [32]
 94 ; both characterise repeatability through RMS/PV quantities or pixel-wise standard-
 95 deviation matrices [31, 32]. ~~Uncertainty-oriented surface metrology embeds,~~
 96 ~~i.e., through amplitude statistics.~~ Pappas et al. propagated the uncertainty of
 97 areal surface-texture parameters ~~in using the metrological-characteristics approach,~~
 98 ~~embedding texture parameters in a GUM-style propagation framework [33].~~ ~~Functional~~
 99 ~~data analysis provides a statistical~~ In statistics, Gertheiss et al. reviewed functional
 100 data analysis as a language for function-valued observations [34, 35], and [34],
 101 and Happ et al. provided a framework separating amplitude and phase variation
 102 in functional data [35]. Scale-dependence has been examined by Wolf, who
 103 connected surface topography with topological description [36], by Sanner et al.,
 104 who defined scale-dependent roughness ~~studies show that characterisation can~~
 105 ~~depend~~ parameters for topography analysis [37], and by Zhao et al., who studied
 106 fractal simulation of topography for lubrication prediction [38]; collectively
 107 these show that surface characterisation depends on bandwidth and filtering
 108 scale [36–38].

109 ~~The research gap is therefore specific — Existing interferometric literature~~
 110 ~~provides robust instrumentation and phase reconstruction; standards-oriented~~
 111 ~~work provides scalar profile and form parameters with uncertainty evaluation ;~~
 112 ~~and functional data analysis provides a mathematical language for function-valued~~
 113 ~~observations. —What has received less attention is a controlled, within-profile~~
 114 ~~comparison of the sensitivity of classical ISO amplitude parameters against~~
 115 ~~variation-based functional norms ;~~ None of these strands, however, resolves a
 116 specific practical problem of profile and roundness metrology. When a localised
 117 geometric feature — a flat, a pit, a scratch — is present on an otherwise smooth

118 artefact, amplitude-averaging parameters such as R_a dilute its signature over
 119 the entire evaluation length, so the feature can remain numerically invisible
 120 although it is metrologically relevant. Existing remedies do not fully address
 121 this for profile data: areal parameters require calibrated height maps that
 122 single-image radial extraction does not provide; visual inspection of polar plots
 123 is operator-dependent and not quantitative; and template-based defect detectors
 124 require the defect geometry to be known in advance. The unresolved question
 125 addressed here is therefore whether gradient-sensitive functional norms — whose
 126 mathematical structure makes them respond to localised slope content rather
 127 than to averaged amplitude — provide reproducible, quantifiable sensitivity to
 128 such features when computed from the same ~~nanometre residual and validated~~
 129 ~~against an external tactile reference~~ reconstructed residual as the standard ISO
 130 parameters, and how large that sensitivity advantage is under controlled conditions.
 131 What has been missing in the literature is precisely this controlled, like-for-like
 132 comparison on identical nanometre residuals with external tactile validation.
 133 The present work addresses this by: (i) injecting synthetic defects of known ge-
 134 ometry into measured profiles to quantify descriptor response under controlled
 135 conditions; (ii) applying the full pipeline to two DAKS-DKD-certified arte-
 136 facts at large scale (24,000 and 100,000 interferograms) with a self-built radial
 137 excess-fraction interferometer; and (iii) benchmarking the optical reconstruction
 138 against 104 tactile reference measurements. Areal parameters are not reported
 139 because the single-image radial extraction does not produce a calibrated areal
 140 height map. The aim is to assess whether L^p , bounded-variation, and Sobolev
 141 seminorms, evaluated alongside ISO ~~4287-21920-2~~ profile parameters on the
 142 same reconstructed residual profile, offer complementary descriptive sensitivity
 143 to controlled profile perturbations, and to evaluate the within-run repeatability
 144 of all quantities across five recorded sweeps. No claim of novelty is attached
 145 to the norms themselves — they are standard tools of functional analysis, and
 146 the functional-analytic vocabulary is used only as an organising taxonomy; the
 147 controlled comparison framework is the contribution.

2. Experimental Details

2.1. Optical Setup

The optical layout of the measurement system is shown in Fig. 1. The system is a two-channel modified Michelson interferometer with channels shifted by approximately 90° in phase. When an array detector is used instead of a point detector, the second channel is not strictly required [39]. In this system it was retained because it is useful for object centring and can serve as a control channel.

The optical setup and the full apparatus-level description were reported in a previous study on radial image processing for phase extraction in rough-surface interferometry [25]. They are repeated here in condensed form to make the present functional-descriptor analysis self-contained and to document the acquisition conditions behind the FN 111 and FN 101 interferogram archives.

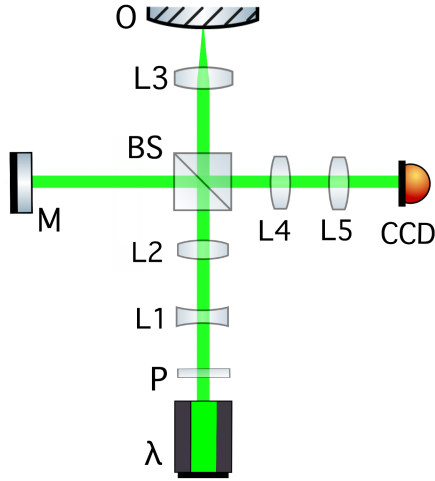


Figure 1: Optical layout of the two-channel modified Michelson interferometer: λ – diode laser (532 nm); P – linear polariser; L – lenses; L_3 – microscope objective; M – reference mirror ($\lambda/10$ flatness); BS – non-polarising beamsplitter; O – object on rotation stage; CCD – camera.

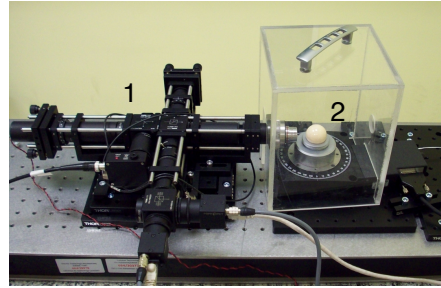


Figure 2: The complete measurement system: (1) modified Michelson interferometer in cage mounting, (2) high-precision rotation stage with the artefact.

161 The system was first tested with the JENOPTIK roundness standard FN 111
 162 (photographs: Supplementary Fig. S1) (Al_2O_3 , $\varnothing = 29.9588$ mm) [40]. A single-
 163 mode 4.5 mW diode laser with wavelength 532 nm was used as the light source.
 164 A current-stabilised supply powered the laser to obtain a stable wavelength [41].
 165 The laser beam passes through a polariser (P in Fig. 1) to set the polarisation
 166 of the light at 45° to the setup plane, minimising the difference in reflectivity
 167 between the two channels.

168 Key components: achromatic lenses $L_{1,2,4,5}$, non-polarising beamsplitter
 169 BS , $\lambda/8$ liquid phase retarder, $\lambda/10$ reference mirror M , DIN-4 microscope
 170 objective L_3 (NA 0.10), polarising beamsplitter PBS , and two Gigabit Ether-
 171 net CCD cameras (640×480 px, 120 fps).

172 The interferometer was built in a cage system for thermal, acoustic and me-
 173 chanical stability. A plexiglass box shields the measurement object, and the
 174 structure is mounted on an area translation stage. A high-precision rotation
 175 stage (encoder accuracy 2.0 mdeg, typical wobble $\pm 12 \mu\text{rad}$, typical eccentric-
 176 ity $\pm 0.40 \mu\text{m}$; full specifications: Supplementary Table S1) rotates the object.
 177 Both interferometer and stage are mounted on a honeycomb breadboard with
 178 vibration isolation.

179 Measurements were performed at night (23:00–05:00) to minimise building-
 180 activity disturbances. The setup was thermally and acoustically shielded (plex-
 181 iglass box, wool blanket) and placed on a granite plate on layered wooden-
 182 polystyrene vibration isolation (photographs: Supplementary Fig. S2). Five
 183 semiconductor temperature sensors ($\pm 0.2^\circ\text{C}$) monitored stability; the tempera-
 184 ture near the reference and object arms remained stable to $\pm 0.2^\circ\text{C}$ over six-hour
 185 acquisitions.

186 2.2. Control System and Acquisition Workflow

187 Cameras were controlled by FLIR SDK-based C programs and Bash scripts [42].
 188 Capture time was recorded per frame for angular-position verification. The ro-
 189 tation stage was controlled via USB ASCII commands [43]; a 25 s settling delay
 190 ensured constant velocity before acquisition began. Object centring was per-

191 formed manually using the two-camera phase-shifted configuration at 45° rota-
192 tion increments. Fringe evaluation used modified open-source R code [39, 44].
193 Table 1 lists the acquisition parameters for both archives; five 400° sweeps were
194 recorded per artefact, with the 40° overlap retained for profile-end comparison.

Table 1: Measurement and sampling parameters of the interferometric acquisition workflow.

Parameter	FN 111	FN 101
Rotation speed, ω	$0.50^\circ/\text{s}$	$0.08^\circ/\text{s}$
CCD frame rate	6 fps	4 fps
Recorded sweeps, n	5	5
Total frames, N	24,000	100,000
Angular step, $\Delta\alpha = \omega/\text{fps}$	$0.083^\circ/\text{frame}$	$0.020^\circ/\text{frame}$
Frames per degree	12	50
Frames per useful 360°	4320	18,000
Frames per recorded 400° sweep	4800	20,000
Overlap per sweep	$40^\circ / 480 \text{ frames}$	$40^\circ / 2000 \text{ frames}$
Total acquisition time	4000 s	25,000 s

195 The object was measured in one continuous five-sweep rotation without repo-
196 sitioning. Each recorded sweep spans 400° ; successive useful 360° segments are
197 shifted by 40° , giving a natural between-sweep angular offset. The 40° overlap
198 region provides a model-free repeatability check: RMS height difference between
199 consecutive sweeps is 12 nm (FN 111) and $0.18 \mu\text{m}$ (FN 101), consistent with
200 Table 3.

201 The frame rates in Table 1 (6 and 4 fps) are far below the 120 fps capability
202 of the cameras and merit comment. The angular step is set by the ratio
203 $\Delta\alpha = \omega/\text{fps}$, so angular resolution is governed jointly by rotation speed and
204 frame rate; measurement time is governed by rotation speed alone. The rotation
205 speeds (0.50 and $0.08^\circ/\text{s}$) were deliberately conservative: slow, constant-velocity
206 rotation minimises dynamic excitation of the stage and keeps the fringe pattern
207 quasi-static within each exposure, and the settling-time protocol (25 s before

acquisition) assumes steady-state velocity. At these speeds, 6 and 4 fps already yield 12 and 50 frames per degree — an oversampling factor of roughly 300 and 1200 relative to the 15 UPR metrological bandwidth — so the frame rate was not the accuracy-limiting factor; increasing it at fixed rotation speed would only add redundant frames. Shorter total acquisition (4,000 and 25,000 s here) would instead require faster rotation with a proportionally higher frame rate, which is technically feasible with the present cameras up to 120 fps. Whether fringe contrast and stage stability are preserved at, say, a tenfold higher speed has not been tested; the trade-off between acquisition speed, vibration robustness, and per-frame phase-extraction quality is identified as a topic for future work, since measurement time is a critical parameter for industrial implementation.

3. Image Processing and Functional-Descriptor Method

3.1. Processing Pipeline

The pipeline follows the radial excess-fraction method of a previous study [25] with added sinusoidal refinement. Each interferogram $I_i(x, y)$ is decoded independently to a wrapped excess fraction ϵ_i ; the sequence is unwrapped chronologically, converted to nanometres ($z_i = \nu_i \lambda / 2$), and reduced by the object-specific LSCI reference model. Sweep boundaries are retained for repeatability analysis.

3.2. Step-by-Step Radial Phase Decoding

For each interferogram $I_i(x, y)$, a radial intensity profile about the fringe centre is extracted and B-spline-smoothed; bright-ring maxima are detected in D^2 space, where circular-fringe theory gives $D_{ij}^2 = c_i(j + \epsilon_i - 1)$ for the j th ring of diameter D_{ij} . A linear fit $D_{ij}^2 = \alpha_i j + \beta_i$ yields the wrapped excess fraction $\epsilon_i = (1 + \beta_i / \alpha_i) \bmod 1$ (2). Where the radial fringe signal permits ($R^2 \geq 0.65$, phase within 0.20 fringe of the ring-diameter estimate), a sinusoidal refinement is applied; otherwise the ring-diameter estimate is retained. The ordered series ϵ_i is unwrapped chronologically to a continuous fringe-order coordinate ν_i , and

the nanometre profile follows from $z_i = \nu_i \lambda / 2$ with $\lambda = 532$ nm (one fringe order $\equiv 266$ nm) (3).

$$D_{ij}^2 = c_i(j + \epsilon_i - 1), \quad (1)$$

$$\epsilon_i = \left(1 + \frac{\beta_i}{\alpha_i}\right) \bmod 1. \quad (2)$$

$$z_i = \nu_i \frac{\lambda}{2}, \quad \lambda = 532 \text{ nm}. \quad (3)$$

3.3. Topography Profile, Reference Removal and Eccentricity Reduction

After phase-to-height conversion, the roundness profile is extracted following ISO 12181-1 [45] (which supersedes the withdrawn ISO 4291). For each 360° segment, a least-squares reference circle (LSCI) is fitted to the angular height profile $z_i = h(\theta_i)$ in Cartesian coordinates:

$$(x_i, y_i) = ((R_0 + z_i) \cos \theta_i, (R_0 + z_i) \sin \theta_i), \quad (4)$$

where $R_0 = 14.9794$ mm is the nominal radius of the JENOPTIK FN 111. The LSCI centre (x_c, y_c) minimises $\sum_i [\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R_0]^2$, yielding the centred radial residuals ρ_i . The LSCI fit is numerically equivalent to removing the first harmonic of the angular profile, but it provides the formal ISO 4291 reference circle required for roundness metrology. After LSCI centring, a phase-correct Gaussian low-pass profile filter with 50% transmission at 15 undulations per revolution (UPR) is applied to the residual profile, consistent with the filter specification used for the tactile reference measurements and with the Gaussian profile filter defined in ISO 16610-21. All classical and gradient-based descriptors are computed from this LSCI-centred, Gaussian-filtered residual $\rho_i^{(\text{filt})}$.

For the JENOPTIK magnification standard FN 101, the same processing chain is applied: a linear trend is removed from the unwrapped phase sequence before height conversion (Sec. 5.1), the LSCI reference circle is fitted to the angular profile, and the Gaussian low-pass filter (50% at 15 UPR) is applied. Although both artefacts are processed through identical filtering steps, the 15 UPR cutoff corresponds to different spatial wavelengths on the circumference: $2\pi \times 14.9794 \text{ mm} / 15 = 6.27 \text{ mm}$ for the FN 111 ($\varnothing = 29.9588 \text{ mm}$) and

261 $2\pi \times 15 \text{ mm}/15 = 6.28 \text{ mm}$ for the FN 101 ($\varnothing \approx 30 \text{ mm}$) — effectively identical.
 262 The angular sampling differs substantially (0.083° vs. 0.020° , Table 1), but the
 263 Gaussian filter sigma is set by the UPR cutoff, not by the sampling interval, so
 264 the filtered profiles are directly comparable.

265 3.4. Classical Profile and Roundness Metrics

266 All classical quantities are evaluated on nanometre residuals, not on raw
 267 intensities, following the profile method of ISO [21920-2 \[46\]](#), [which supersedes](#)
 268 [the earlier ISO 4287 \[47\]](#); [the amplitude-parameter definitions used here are](#)
 269 [common to both documents, and ISO 4287 is retained as a secondary reference](#)
 270 [because the tactile comparison data and the previous study \[25\] predate the](#)
 271 [transition](#). Areal parameters (ISO 25178 [48]) are not reported because the
 272 single-image radial extraction does not produce a calibrated areal height map.
 273 For a residual profile or vector of residual pixels p , the reported profile-style
 274 roughness descriptors are

$$R_a = \frac{1}{M} \sum_{j=1}^M |p_j - \bar{p}|, \quad R_q = \left[\frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^2 \right]^{1/2}, \quad (5)$$

275 and

$$R_z = \max_j (p_j - \bar{p}) - \min_j (p_j - \bar{p}). \quad (6)$$

276 [As implemented, this quantity is the global peak-to-valley height of the full](#)
 277 [evaluation profile, corresponding to the total-height parameter \(\$R_t\$ in ISO 21920-2](#)
 278 [terms\) rather than the section-averaged \$R_z\$; the symbol \$R_z\$ is retained for](#)
 279 [continuity with the previous study \[25\]. This implementation detail is relevant](#)
 280 [to the interpretation of Table 3 \(Sec. 4.3\).](#) The profile skewness R_{sk} and kurtosis
 281 R_{ku} — ISO [4287-21920-2](#) shape parameters that characterise asymmetry and
 282 peakedness of the height distribution — are also reported:

$$R_{sk} = \frac{1}{R_q^3} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^3, \quad R_{ku} = \frac{1}{R_q^4} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^4. \quad (7)$$

283 $R_{sk} = 0$ for a symmetric distribution; $R_{sk} < 0$ indicates a predominance
 284 of valleys over peaks. $R_{ku} = 3$ for a normal distribution; $R_{ku} > 3$ indi-
 285 cates a spikier (leptokurtic) profile. These are the only ISO [4287-21920-2](#)

286 [amplitude-distribution](#) parameters that capture distribution shape beyond am-
 287 plitude averages, and their inclusion allows a direct test of whether gradient-
 288 based descriptors provide information beyond even the best ISO shape param-
 289 eters. For the eccentricity-corrected ceramic angular profile $q_C(\theta)$,

$$RONt = \max_{\theta} q_C(\theta) - \min_{\theta} q_C(\theta), \quad RONq = \left[\frac{1}{2\pi} \int_0^{2\pi} q_C^2(\theta) d\theta \right]^{1/2}. \quad (8)$$

290 These are profile/form quantities in nanometres. B-spline pre-smoothing (20 de-
 291 grees of freedom, $\approx 18^\circ$ period) stabilises the LSCI fit; the subsequent Gaussian
 292 low-pass filter (50% at 15 UPR, ISO 16610-21) defines the metrological band-
 293 width. All descriptors are computed from the same Gaussian-filtered residual.

294 3.5. Functional Gradient-Based Descriptors

295 The concept of a complete normed vector space originates from the founda-
 296 tional work of Banach [49]. For discrete profile data of finite length N , the vector
 297 space $\mathbb{R}^N$ equipped with any norm is a Banach space; the present work does
 298 not invoke infinite-dimensional completeness but draws on the taxonomy of L^p ,
 299 Sobolev, and bounded-variation norms that Banach's framework systematised.
 300 For a centred residual profile $p(s)$ and $1 \leq p < \infty$,

$$\|p\|_p = \left(\frac{1}{L} \int_0^L |p(s)|^p ds \right)^{1/p}, \quad \|p\|_\infty = \max_{s \in [0, L]} |p(s)|. \quad (9)$$

301 The reported functional descriptors include L^1 , L^2 , L^∞ , the one-dimensional
 302 Sobolev $W^{1,2}$ seminorm and the bounded-variation (BV) total variation,

$$|p|_{W^{1,2}} = \left(\int_0^L \left| \frac{dp}{ds} \right|^2 ds \right)^{1/2}, \quad TV(p) = \int_0^L \left| \frac{dp}{ds} \right| ds. \quad (10)$$

303 In discrete form, for a uniformly sampled profile p_j ($j = 1, \dots, N$) with angular
 304 spacing $\Delta\alpha$,

$$|p|_{W^{1,2}} \approx \left(\Delta\alpha \sum_{j=1}^{N-1} \left(\frac{p_{j+1} - p_j}{\Delta\alpha} \right)^2 \right)^{1/2}, \quad TV(p) \approx \sum_{j=1}^{N-1} |p_{j+1} - p_j|. \quad (11)$$

305 Both $W^{1,2}$ and BV density carry sampling-interval dependence ~~and require conversion~~
 306 ~~to angular units ($\text{nm deg}^{-1/2}$ and nm deg^{-1} , respectively) for cross-instrument~~

307 ~~comparison~~; “nm sample⁻¹” is retained in the result tables for the fixed-acquisition
 308 setup, ~~and a standardised angular-unit convention for cross-instrument comparison~~
 309 is defined in Sec. 3.6.

310 The curvature RMS (root-mean-square of the discrete second derivative;
 311 reported in Supplementary Table S2) carries a $(\Delta\alpha)^{-2}$ sampling dependence and
 312 should not be compared across instruments with different angular resolution.

313 3.6. ~~Repeatability Uncertainty~~Angular-Unit Convention for Sampling-Dependent 314 Descriptors

315 Gradient-based descriptors evaluated on discretely sampled profiles depend
 316 on the sampling interval: the discrete total variation of an unfiltered profile
 317 grows as the sampling grid refines (noise contributes gradient content at every
 318 scale), so raw per-sample values are instrument-specific. For meaningful cross-instrument
 319 comparison the following convention is proposed and used here.

- 320 1. **Band-limit first.** The profile is filtered with a declared phase-correct
 321 Gaussian profile filter (ISO 16610-21) at a stated cutoff in undulations
 322 per revolution (here 50% transmission at 15 UPR). After band-limiting,
 323 the profile derivative is bounded and the discrete sums converge to the
 324 continuous integrals as $\Delta\alpha \rightarrow 0$; the descriptor then characterises the
 325 declared bandwidth rather than the sampling grid.
- 326 2. **Sample adequately.** The angular sampling must resolve the declared
 327 bandwidth with margin; at least ten samples per cutoff undulation are
 328 recommended. Here the FN 111 and FN 101 profiles carry ≈ 290 and
 329 ≈ 1200 samples per 15-UPR undulation, so discretisation error is negligible.
 330
- 331 3. **Report in angular units.** The BV density is reported as total variation
 332 per unit angle, $BV_\theta = TV(p)/\Theta$ in nm deg⁻¹ ($\Theta = 360^\circ$), and the gradient
 333 content as the RMS angular gradient $g_{\text{rms}} = [\frac{1}{N-1} \sum_j ((p_{j+1} - p_j)/\Delta\alpha)^2]^{1/2}$
 334 in nm deg⁻¹, together with the filter cutoff. For comparison across artefacts
 335 or instruments of different diameter, the angular unit is additionally converted

to an arc-length unit by dividing by the arc length per degree, $\pi D/360$; the BV density is then $BV_s = TV(p)/(\pi D)$ in nm mm^{-1} . The arc-length form is the diameter-independent unit and is the one recommended for cross-instrument reporting, since nm deg^{-1} still scales with the circumference of the specific artefact.

A note on physical dimensions clarifies why this is necessary and is itself part of the answer to the redundancy question (Sec. 5.3). Amplitude descriptors (R_a , R_q , L^1 , L^2 , L^∞ , $RONt$) have the dimension of length (nm) and are sampling-independent. Gradient descriptors are dimensionally distinct: BV density and the RMS gradient are height-per-length (nm mm^{-1}), i.e. dimensionless slopes, while the $W^{1,2}$ seminorm as defined here is $\text{height} \times \text{length}^{-1/2}$. This dimensional separation is the reason the gradient layer carries information that no amplitude parameter can encode, and also the reason its raw per-sample value must be qualified by a sampling and filter declaration.

For the present data, the per-sample values in Table 3 convert as follows. FN 111 ($\Delta\alpha = 0.083^\circ$): BV density $0.263 \text{ nm sample}^{-1} \rightarrow 3.2 \text{ nm deg}^{-1} \rightarrow 12.1 \text{ nm mm}^{-1}$; RMS gradient $0.310 \text{ nm sample}^{-1} \rightarrow 3.7 \text{ nm deg}^{-1} \rightarrow 14.2 \text{ nm mm}^{-1}$. FN 101 ($\Delta\alpha = 0.020^\circ$): BV density $6.58 \text{ nm sample}^{-1} \rightarrow 329 \text{ nm deg}^{-1} \rightarrow 1.26 \text{ } \mu\text{m mm}^{-1}$; RMS gradient $8.76 \text{ nm sample}^{-1} \rightarrow 438 \text{ nm deg}^{-1} \rightarrow 1.67 \text{ } \mu\text{m mm}^{-1}$, all at the 15-UPR cutoff. The two artefacts have almost identical circumference ($\pi D \approx 94.1$ and 94.2 mm), so the large difference in arc-length gradient content (12 versus 1257 nm mm^{-1}) reflects a genuine difference in surface variation rather than a unit artefact. Because each conversion is a fixed multiplicative factor for a given acquisition, relative uncertainties and relative defect responses are identical in every unit system. Adoption of gradient descriptors in reference software for roundness evaluation would require exactly this pair of declarations — filter cutoff and arc-length units — which parallels the established practice of declaring the filter for $RONt$.

3.7. Uncertainty Evaluation

Each dataset contains five 400° sweeps recorded in a single continuous rotation without stopping or repositioning. The five useful 360° segments are extracted from this continuous sequence; because each recorded sweep spans 400° , successive useful revolutions are shifted by 40° on the object. These segments are not strictly independent — they share the same continuous-run conditions (laser warm-up state, stage-bearing temperature, ambient drift). The standard deviation across sweeps therefore reflects within-run repeatability rather than between-run reproducibility. For a descriptor X evaluated on each sweep, the mean value is $\bar{X}$ and the within-run standard uncertainty is estimated as

$$u_{\text{within}}(X) = \frac{s_X}{\sqrt{n}}, \quad n = 5, \quad (12)$$

where s_X is the sample standard deviation across the five sweeps. The reported expanded uncertainty is

$$U_{95}(X) = t_{0.975, n-1} u_{\text{within}}(X). \quad (13)$$

With $n = 5$, $t_{0.975, 4} = 2.776$; the χ^2_4 -based 95% CI for the standard deviation spans $[0.6s_X, 2.9s_X]$, so U_{95} is indicative rather than definitive. Because the five segments share continuous-run conditions (laser state, stage temperature, ambient drift), the $1/\sqrt{n}$ factor in u_{within} may underestimate the true uncertainty of the mean; all reported uncertainties are *lower bounds*. ~~A full GUM-compliant budget (An indicative type-B terms: laser wavelength stability, refractive-index variations, Abbe error [50], camera nonlinearity, form-removal uncertainty) is beyond the present scope~~ budget for the principal systematic terms is given in Sec. 3.8; a complete GUM-compliant budget [51] with experimentally characterised input distributions, and a between-run reproducibility study with independent setups, thermal cycles, and physical re-centring of the artefact, remain future work (Sec. 5.6). With $n = 5$ sweeps and the observed within-run variability, the minimum detectable relative difference between two descriptors (80% power, $\alpha = 0.05$, two-sided paired t -test) is approximately 30% for the FN 111 and

5% for the FN 101, consistent with the exploratory character of the descriptor comparison. A per-sweep trend of FN 111 $RONt$ values gives a slope of $8.4 \pm 4.1 \text{ nm sweep}^{-1}$ ($p = 0.12$), consistent with zero drift, though the sweep-to-sweep scatter cautions that U_{95} may underestimate long-term reproducibility.

3.8. *Indicative Type-B Uncertainty Budget*

Although a complete GUM budget requires between-run data that the present single-rotation acquisition cannot supply, the principal type-B contributions can be bounded from component specifications and pipeline properties. Table 2 summarises the estimates for the FN 111 $RONt$ evaluation; the same relative arguments apply to the FN 101.

Laser wavelength stability. The height scale $z = \nu\lambda/2$ inherits the relative wavelength error: a $\pm 1 \text{ nm}$ tolerance on the 532 nm diode module (rectangular distribution) gives a relative standard uncertainty of 0.11%, i.e. 0.3 nm on $RONt = 278 \text{ nm}$ (and 46 nm on the FN 101 $RONt = 42.07 \mu\text{m}$, within its $\pm 0.48 \mu\text{m}$ repeatability interval). Wavelength error acts as a pure scale factor and cannot create or mask localised features.

Refractive index of air. The in-air wavelength varies with temperature and pressure by $\sim 10^{-6}$ per 1°C or 3 hPa; with the observed $\pm 0.2^\circ\text{C}$ stability the contribution is below 10^{-5} relative, i.e. $< 0.01 \text{ nm}$. Negligible.

Abbe error and stage wobble. The specified axis wobble ($\pm 12 \mu\text{rad}$) enters through two mechanisms: a cosine-type radial error $R\theta_w^2/2 \approx 1 \text{ pm}$, and a lateral beam walk $R\theta_w \approx 0.18 \mu\text{m}$ on the circumference, equivalent to an angular mislocation of 0.0007° , which maps through the local band-limited gradient ($\approx 3.2 \text{ nm deg}^{-1}$, Sec. 3.6) to $\approx 0.002 \text{ nm}$. Both are negligible after LSCI first-harmonic removal.

Camera nonlinearity and EFM localisation. Intensity nonlinearity perturbs the radial fringe-peak positions; its effect is bounded empirically by the per-frame excess-fraction uncertainty of ± 0.05 fringe (13 nm). The 15-UPR band limit averages ≈ 290 frames per resolved undulation, reducing the per-point contribution to $\approx 0.8 \text{ nm}$ and the peak-to-valley contribution to $\approx 1.1 \text{ nm}$.

420 *Form removal.* The LSCI reference (first-harmonic removal) is exact only in
421 the limaçon approximation; the second-order term $a^2/(2R)$ for residual centring
422 eccentricity a leaks into the second harmonic of the residual. Manual centring
423 leaves a of the order of tens of micrometres (the shared first-harmonic model
424 of Sec. 5.2.1, with its 63 nm/0.18% per-sweep amplitude variation, implies
425 $a \approx 35 \mu\text{m}$); for $a = 15\text{--}35 \mu\text{m}$ the term evaluates to 7–41 nm. Because this
426 contribution is common to all five sweeps, it does not appear in the type-A
427 scatter and must be carried as a bounded systematic term. Its magnitude is
428 consistent with the 34 nm difference between the optical *RONt* (278 ± 22 nm)
429 and the certified tactile value (244 ± 52 nm), a difference that the $E_n = 0.61$
430 comparison shows to be within the combined uncertainties.

Table 2: Indicative type-B uncertainty budget for the FN 111 *RONt* evaluation. Random contributions are standard uncertainties combined in quadrature with the type-A term; the form-removal term is a bounded systematic (bias) contribution common to all sweeps and is reported separately.

Source	Basis	Contribution to <i>RONt</i>
Within-run repeatability (type A)	$s/\sqrt{5}$, five sweeps	7.9 nm
Laser wavelength (± 1 nm, rect.)	module specification	0.3 nm
Air refractive index ($\pm 0.2^\circ\text{C}$)	Edlén-type sensitivity	< 0.01 nm
Abbe error / axis wobble ($\pm 12 \mu\text{rad}$)	stage specification, geometry	< 0.01 nm
Camera nonlinearity / EFM localisation	± 0.05 fringe per frame, band-limited averaging	1.1 nm
Combined standard uncertainty u_c (random)	quadrature	8.0 nm
Expanded U_{95} (random, $t_{0.975,4}$)		22 nm
Form removal (limaçon second order)	$a^2/(2R)$, $a = 15\text{--}35 \mu\text{m}$	≤ 41 nm (syst. bound)

431 The quadrature combination is dominated by the type-A term, so the reported
432 U_{95} values are essentially unchanged by the enumerated random type-B terms;
433 the budget nevertheless shows that the leading unquantified risks are the form-removal
434 systematic and the untested between-run effects, not the enumerated instrument
435 terms.

436 3.9. Computational Implementation

437 The reconstruction workflow (`scripts/reconstruct_radial_profile_sequence.py`)
438 decodes ϵ_i from each interferogram, unwraps the sequence, converts to
439 nanometres, applies LSCI fitting and Gaussian low-pass filtering (50% at 15 UPR,
440 ISO 16610-21), and exports CSV/JSON metrics with sweep-repeatability uncer-
441 tainty. Vector PDF figures document each processing stage. The pipeline is in-
442 voked as `PYTHONPATH=src.venv/bin/pythonscripts/reconstruct_radial_profile_`
443 `sequence.py` with dataset-specific parameters (full command in the data repos-
444 itory). Per-image excess-fraction decoding and unwrapping require approxi-
445 mately 0.5 ms per frame (measured on an Intel Core i7-9700K, single-threaded
446 Python 3.9), so the full 124,000-frame archive processes in under two minutes
447 on consumer hardware.

448 4. Results

449 The results follow the updated radial sequence route only. Each raw im-
450 age is first decoded to a wrapped excess fraction from its radial fringe pat-
451 tern. The ordered excess-fraction series is then unwrapped in chronological
452 order, converted to nanometres, and corrected for eccentricity. Finally, classi-
453 cal profile/roundness descriptors and functional gradient-based descriptors are
454 computed from the same nanometre residual profiles.

455 4.1. Radial Phase Decoding and Nanometre Profile Reconstruction

456 The FN 111 folder contained 24,000 interferograms and the steel folder con-
457 tained 100,000 interferograms. The reported run used full-resolution images, the

central radial aperture of 240 px, B-spline-smoothed peak detection, continuity-aware EFM peak-subset tracking and an EFM prominence threshold of 0.04 of the radial fringe-signal range. All five recorded sweeps were processed.

Figs. S3–S6 (single-image EFM and sequence processing steps) are provided in the Supplementary Materials.

4.2. *Roughness, Roundness and Functional Descriptors*

Table 3 gives the numerical comparison after LSCI reference circle and Gaussian low-pass filtering (50% at 15 UPR). Both artefacts are processed through the same filtering chain. Each descriptor is reported as the full-sequence value together with the 95% expanded type-A repeatability uncertainty estimated from the five sweeps. Per-sweep values for all descriptors are provided in Supplementary Table S3.

~~The numerical identity $RONt = R_z$~~

4.3. *Coincidence of R_z with $RONt$ and R_q with $RONq$*

~~The identical values of R_z and $RONt$ (and of R_q and $RONq$) for both artefacts (in Table 3) reflects the fact that after LSCI centring and Gaussian low-pass filtering (50% at 15 UPR), the residual profile is dominated by a single lobe (harmonic decomposition are expected by construction, not coincidental. In this pipeline both descriptor families are evaluated on the same LSCI-centred, Gaussian-filtered closed residual profile (Sec. 3.3; harmonic decomposition of the residual: Supplementary Fig. S19); the roughness-style R_z is the global peak-to-valley height of that profile (i.e., the total height Rt ; see the implementation note in the definition), while $RONt$ is the peak-to-valley roundness deviation and the profile roughness peak-to-valley therefore coincide. This identity is specific to the present artefact-filter combination and should not be expected to hold for artefacts with multi-lobed form deviations or for different filter cutoffs of the very same profile, so the two coincide exactly; the RMS pair $R_q/RONq$ coincides for the same reason. The two families would separate if profile texture were evaluated on a different bandwidth than form (e.g., after~~

Table 3: Full-resolution radial EFM reconstruction from raw interferograms (LSCI + Gaussian low-pass, 50% at 15 UPR, ISO 16610-21). Uncertainties are 95% expanded within-run repeatability estimates from five sweeps ($n = 5$, $t_{0.975,4} = 2.776$). Units: nm for all descriptors except BV density and $W^{1,2}$ gradient (nm sample⁻¹). Certified *RONt*: FN 111 244 ± 52 nm (Hommel Roundscan 535, $n = 104$); FN 101 $41.94 \mu\text{m}$ (form tester, $n = 12$). The FN 101 certificate does not report an expanded uncertainty; therefore only the percentage difference is reported for this artefact rather than an E_n score.

Descriptor	FN 111	FN 101
R_a	66.3 ± 4.7 (<u>66.3 ± 4.7</u>) nm	8.525 ± 0.115 (<u>8.525 ± 0.115</u>) μm
R_q	77.6 ± 5.0 (<u>77.6 ± 5.0</u>) nm	10.376 ± 0.090 (<u>10.376 ± 0.090</u>) μm
R_z	278 ± 22 (<u>278 ± 22</u>) nm	42.07 ± 0.48 (<u>42.07 ± 0.48</u>) μm
<i>RONt</i>	278 ± 22 (<u>278 ± 22</u>) nm	42.07 ± 0.48 (<u>42.07 ± 0.48</u>) μm
<i>RONq</i>	77.6 ± 5.0 (<u>77.6 ± 5.0</u>) nm	10.376 ± 0.090 (<u>10.376 ± 0.090</u>) μm
L^1 mean	66.3 ± 4.7 (<u>66.3 ± 4.7</u>) nm	8.525 ± 0.115 (<u>8.525 ± 0.115</u>) μm
L^2 RMS	77.6 ± 5.0 (<u>77.6 ± 5.0</u>) nm	10.376 ± 0.090 (<u>10.376 ± 0.090</u>) μm
L^∞	150 ± 12 (<u>150 ± 12</u>) nm	21.97 ± 0.42 (<u>21.97 ± 0.42</u>) μm
BV total	1.137 ± 0.062 (<u>1.137 ± 0.062</u>) μm	118.4 ± 6.2 (<u>118.4 ± 6.2</u>) μm
BV density	0.263 ± 0.014 (<u>0.263 ± 0.014</u>) nm sample ⁻¹	6.58 ± 0.35 (<u>6.58 ± 0.35</u>) nm sample ⁻¹
$W^{1,2}$ grad.	0.310 ± 0.015 (<u>0.310 ± 0.015</u>) nm sample ⁻¹	8.76 ± 0.43 (<u>8.76 ± 0.43</u>) nm sample ⁻¹
<i>ISO shape parameters</i>		
R_{sk} (skewness)	-0.09 ± 0.04	0.00 ± 0.09
R_{ku} (kurtosis)	1.98 ± 0.21	2.51 ± 0.08
<i>Derived norm ratios</i>		
L^4/L^1 peakiness	1.39 ± 0.08	1.53 ± 0.03
<i>External validation</i>		
E_n score	0.61	< 1 (0.3% diff.; cf. Fig. S12)

487 λ c-based separation of roughness from waviness in the ISO 21920 sense) or on an
 488 independent linear trace. Both are nevertheless reported because they belong
 489 to different ISO frameworks (profile texture versus roundness) with different
 490 reference conventions, and because their exact equality serves as a verification
 491 that all descriptors share one evaluation profile — the design premise of the
 492 like-for-like comparison.

493 4.4. Classical-Functional-Descriptor Comparison

494 The ~~numerical-identity~~ exact equalities $R_a = L^1$ and $R_q = L^2$ ~~(in Table 3)~~
 495 ~~confirms~~ are definitional — each pair is the same formula applied to the same
 496 centred residual — and are reported purely as a computational cross-check that
 497 both descriptor families are evaluated on ~~the same nanometre residual~~ identical
 498 data, not as an empirical finding (see Sec. 5.3). For the FN 111, BV total
 499 variation carries 1.2% relative expanded uncertainty versus 5.5% for R_a and
 500 8.5% for R_z , suggesting that gradient-based descriptors may exhibit metrological
 501 stability comparable to or better than peak-referenced roughness parameters
 502 in the tested datasets. Comparative bar charts and uncertainty analyses are
 503 provided in the Supplementary Materials (Figs. S7–S8). Ceramic and steel
 504 profile reconstruction plots are shown in Supplementary Figs. S9–S10.

505 The steel roundness polar representation (FN 101, Flick flat) is provided in
 506 the Supplementary Materials (Fig. S11).

JENOPTIK FN~111 — Roundness deviation
 $RONt = 278$ nm (5 sweeps, native angular coordinates)

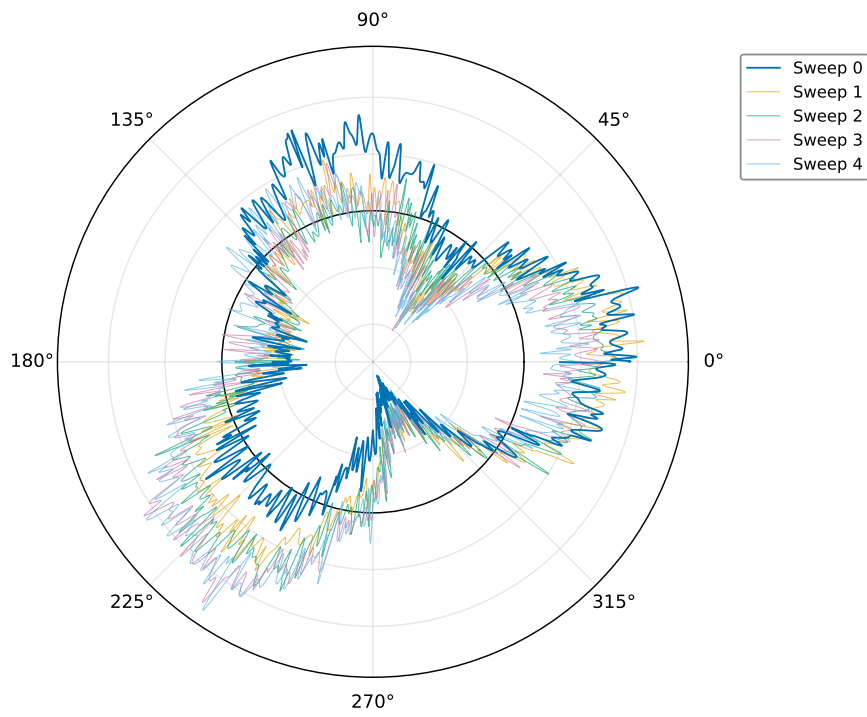


Figure 3: Ceramic roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native positions. $RONt = 278 \pm 22$ nm.

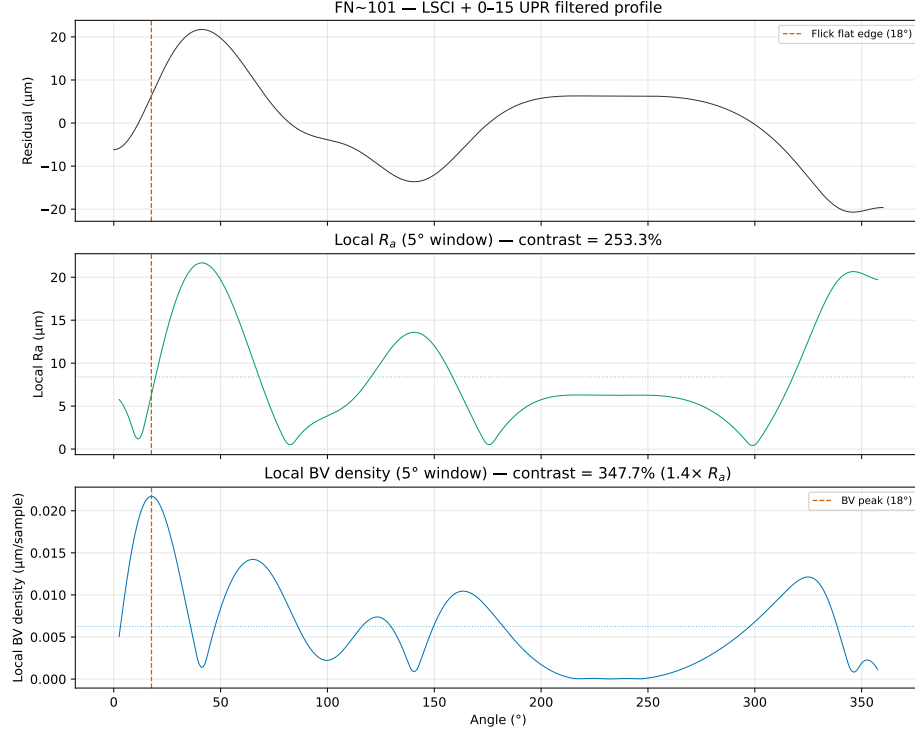


Figure 4: Flick flat detection on the FN 101 magnification standard. BV density (bottom) shows a clear spike at the Flick flat angular position, whereas R_a computed in a sliding window (top) shows no detectable response at the reconstruction resolution. The sliding-window width was matched to the certified angular extent of the Flick flat as specified in the DAkKS-DKD certificate (Art. 532 528); identical window parameters were used for both descriptors.

507 The Flick flat produces a clear response in BV density while showing no
 508 detectable response in a sliding-window R_a analysis at the reconstruction reso-
 509 lution (Fig. 4), illustrating the potential complementary sensitivity of gradient-
 510 based descriptors to localised geometric features.

511 5. Discussion

512 5.1. Confirmed Results

513 The analysis rests on a key metrological distinction: an interferogram is
 514 an intensity measurement, not a phase map and not a height map. The half-

wavelength relation $z = \nu\lambda/2$ becomes valid only after each image has been decoded to a wrapped excess fraction and the ordered excess-fraction sequence has been unwrapped to the continuous fringe-order coordinate ν . This is why the reported comparison uses only reconstructed nanometre profiles, not raw greyscale images and not areal height maps.

The most sensitive part of the workflow is the EFM localisation of bright-ring maxima, with an excess-fraction uncertainty of approximately ± 0.05 fringe orders (tightened to ± 0.03 by sinusoidal refinement in 72% of FN 111 frames). A single ϵ_i error of 0.05 fringe corresponds to 13 nm, sub-dominant to the 85 nm residual-profile RMS. The chronological unwrapping criterion rejects phase jumps exceeding 0.35 fringe. The median number of detected rings is three for both artefacts within the 240 px central aperture. The five 400° ceramic sweeps require a geometric correction: successive useful 360° intervals are shifted by 40° because the physical object angle advances continuously; without this correction, the LSCI fit appears inconsistent even when phase extraction within each sweep is repeatable.

The linear drift of $0.0246 \text{ fringe frame}^{-1}$ in the steel unwrapped phase sequence (versus 0.00021 for the FN 111) is attributed to cumulative bias in the global unwrapping algorithm rather than thermal effects or axial runout [50]. The lower fringe contrast of the FN 101 interferograms — likely due to surface finish — reduces per-frame EFM quality; the pipeline applies automatic linear detrending where needed.

5.2. Interpretation

5.2.1. Eccentricity removal and wobble verification

Roundness descriptors are computed after LSCI reference-circle fitting (Sec. 3.3), which removes the first harmonic. A shared model across all five sweeps yields $R^2 = 0.99991$ with per-sweep amplitude variation of 63 nm (0.18%); the manufacturer-specified wobble ($\pm 12 \mu\text{rad}$) bounds the observed variation an order of magnitude below $RONt = 278 \text{ nm}$. The polar representation (Fig. 3) shows five

overlaid sweeps repeatable to ± 50 nm, illustrating that the residual is dominated by low-order form deviation rather than random roughness.

5.3. Comparison with ISO Parameters

A central question of this work is whether gradient-based functional descriptors provide additional descriptive sensitivity alongside classical ISO parameters when applied to the same reconstructed profiles. To answer this rigorously, every descriptor was computed on the same reconstructed nanometre residual profiles.

Mathematical equivalences. The L^1 norm is defined as $\frac{1}{N} \sum |p_i - \bar{p}|$, which is identical to the definition of R_a . The L^2 norm is $\sqrt{\frac{1}{N} \sum (p_i - \bar{p})^2}$, identical to R_q . Computing both on the same centred residual yields *exact numerical equality*: $R_a - L^1 = 0$ and $R_q - L^2 = 0$ to machine precision for both artefacts (Table 3). These equalities are not empirical findings; they are definitional consequences. Labelling the arithmetic mean of absolute deviations as L^1 rather than R_a provides no new metrological information and should not be presented as a methodological advance.

It is emphasised that neither the L^p norms nor the Sobolev and BV semi-norms are presented as novel mathematical constructs; they are standard tools whose definitions are restated for completeness. Their value in the present context is threefold. First, the functional-analytic taxonomy places ISO parameters at specific positions (L^1 , L^2 , L^∞) within a larger space of possible descriptors, revealing that descriptors outside the ISO set — BV density, $W^{1,2}$ seminorm — respond differently to the same perturbations. Second, the synthetic-defect injection methodology provides a controlled, like-for-like comparison framework that can be applied to any descriptor computed from the same residual profile; the framework itself, rather than any individual norm, constitutes the methodological contribution. Third, the Flick flat on the certified FN 101 standard provides a concrete existence proof: a real, DAKS-DKD-certified artefact whose defining geometric feature is detected unambiguously by BV density while remaining below the detection threshold of sliding-window R_a (Fig. 4).

Extrema: R_z versus L^∞ . $R_z/L^\infty \approx 2$ for both artefacts, consistent with

574 symmetric positive and negative excursions. L^∞ captures the single largest
 575 one-sided excursion; the ratio serves as a simple symmetry indicator.

576 **Variation-based descriptors.** A controlled sensitivity analysis (Supple-
 577 mentary Table S2) examined five synthetic defect types injected into the FN 111
 578 residual profile. Amplitude-averaging descriptors (R_a , R_q , R_z) showed no de-
 579 tectable change at the reconstruction resolution — a single 100 nm spike changed
 580 R_a by less than 0.05%, below the per-frame phase-extraction precision — while
 581 BV density increased by +18% and the $W^{1,2}$ gradient seminorm by +622% for
 582 the same spike (Fig. 5). ISO shape parameters responded to asymmetry (R_{sk}
 583 changed by −1104% for a step) but not to symmetric localised features. R_{sk}
 584 and BV density appear complementary in the tested scenarios: R_{sk} responds to
 585 asymmetry, BV density responds to localised gradients.

586 Translation of synthetic responses to realistic defects. The injected
 587 defects are idealised (isolated spike, uniform scratch, pure sinusoid), whereas
 588 real manufacturing defects have complex, random morphologies. The synthetic
 589 responses nevertheless translate to other backgrounds through a simple scaling
 590 argument. A localised defect adds a total-variation increment ΔTV that depends
 591 only on the defect geometry ($\Delta TV \approx 2A$ for an isolated spike of amplitude
 592 A), so the relative full-profile BV-density response is $\Delta TV/TV_{\text{background}}$. On
 593 the FN 111 background ($TV = 1.137 \mu\text{m}$) a 100 nm spike yields +18%; on
 594 the FN 101 background ($TV = 118.4 \mu\text{m}$) the identical spike would yield only
 595 +0.17%, below the within-run repeatability. Detectability of a given absolute
 596 defect in the full-profile descriptor therefore scales inversely with the background
 597 gradient content, which quantifies the reviewer-relevant observation that the
 598 FN 101 defect-to-background ratio is roughly 500 times less favourable than
 599 typical smooth-surface applications. Two consequences follow for industrial
 600 inspection. First, full-profile BV density is best suited to smooth, finished
 601 surfaces — bearing raceways, optical components, precision seals — whose
 602 residual amplitudes (tens of nanometres) are comparable to the FN 111 case,
 603 and where the synthetic archetypes map onto realistic features (spike $\leftrightarrow$ pit
 604 or inclusion edge, scratch $\leftrightarrow$ handling or tooling mark, sinusoid $\leftrightarrow$ chatter

or vibration signature). Second, for rougher backgrounds the sliding-window formulation restores local contrast: the certified Flick flat is detected on the FN 101 (Fig. 4) despite the unfavourable full-profile ratio, precisely because the window localises the comparison. Validation against physical defects of calibrated dimensions on real metallic and ceramic surfaces remains necessary before these scaling arguments can be considered established (Sec. 5.6).

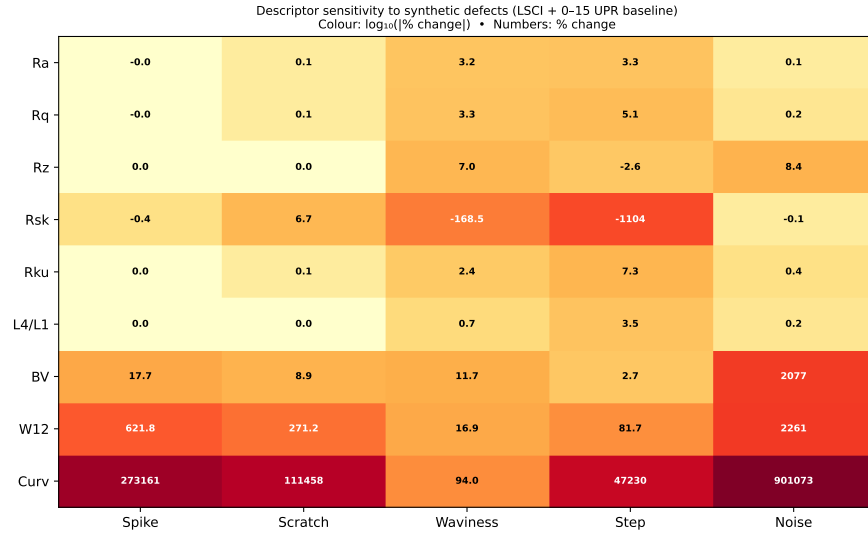


Figure 5: Sensitivity of ISO and gradient-based descriptors to five synthetic defect types injected into the FN 111 residual profile. Percentage change from the unperturbed baseline; bold labels indicate changes exceeding 10%.

Supporting analyses — descriptor correlation matrices, power spectral density comparison, Cohen’s d effect sizes, critical descriptor assessment, and L^p norm convergence — are provided in the Supplementary Materials (Figs. S13–S17). Given the number of descriptors compared (~ 15 across two artefacts), the reported differences should be interpreted as exploratory rather than confirmatory; no formal multiple-comparison correction has been applied.

Practical considerations. A practical suggestion for interferometric profile and roundness metrology is to retain R_a , R_q , R_z , R_{sk} , $RONt$ and $RONq$ as the primary engineering layer compliant with ISO 4287:2019-2, and to consider

620 supplementing them with BV density (in angular units, nm deg^{-1} , [Sec. 3.6](#)) as
 621 a gradient-sensitive functional-analysis complement. The $W^{1,2}$ gradient semi-
 622 norm is [strongly](#) correlated with BV density in the present datasets (within the
 623 FN 111: Pearson $r = 0.95$, $n = 5$ sweeps; within the FN 101: $r = 0.93$, $n = 5$).
 624 The pooled $r = 0.94$ (95% CI $[0.72, 0.99]$, $n = 10$) is reported for descriptive pur-
 625 poses only — the sweeps are not independent observations and this pooled corre-
 626 lation should be interpreted with caution. [This strong correlation indicates that,](#)
 627 [as repeatability metrics across sweeps, BV density and the \$W^{1,2}\$ may add limited](#)
 628 [independent information beyond BV density alone given the small sample. The](#)
 629 [seminorm carry largely redundant information, and routine reporting of both is](#)
 630 [not warranted. They are strongly correlated but not mathematically equivalent:](#)
 631 [BV integrates the first power of the local slope, \$TV = \int |p'| ds\$, whereas \$W^{1,2}\$](#)
 632 [integrates its square, \$\int |p'|^2 ds\$, so the two coincide only for profiles whose slope](#)
 633 [magnitude is nearly constant and diverge whenever the slope distribution is](#)
 634 [uneven. This is exactly what the defect responses show: for the 100 nm](#)
 635 [synthetic spike the \$W^{1,2}\$ seminorm responded with +622% versus +18% for](#)
 636 [BV density, because the quadratic weighting amplifies a single steep event far](#)
 637 [more strongly than the linear total-variation measure. A practical reading of](#)
 638 [the descriptor space in these data is therefore: \(i\) \$R_a/R_q\$ \(equivalently \$L^1/L^2\$ \)](#)
 639 [carry amplitude information; \(ii\) \$R_{sk}\$ carries an independent asymmetry axis;](#)
 640 [\(iii\) BV density carries the localised-gradient axis and is recommended as the](#)
 641 [primary gradient descriptor; \(iv\) the \$W^{1,2}\$ seminorm adds value mainly as a](#)
 642 [high-sensitivity alarm for isolated steep features rather than as an independent](#)
 643 [information axis; and \(v\) the \$L^4/L^1\$ peakiness ratio provides a supplementary](#)
 644 distribution-shape axis with a direct physical interpretation as the amplification
 645 of dominant profile excursions relative to R_a . These additional descriptors are
 646 computed from the same residual profile without additional hardware or changes
 647 to the EFM pipeline. The cross-instrument validation suggests that gradient-
 648 based descriptors are applicable to any 1D profile, whether from interferometry
 649 or tactile roundness measurement.

650 [Practical value and the known-feature caveat. It must be acknowledged](#)

that the Flick flat is a certified, deliberately manufactured feature whose presence and position were known in advance: Fig. 4 is an existence proof under ground-truth conditions, not a blind detection study. The practical argument for a gradient-sensitive layer is therefore not that it revealed an unknown defect here. It is that it converts what an operator might notice in a visual polar-plot inspection into a quantitative, thresholdable, and automatable indicator computed from data already acquired for ISO reporting — with no operator in the loop, a defined angular position estimate, and a repeatability statement. Whether this indicator reveals defects that routine quality-control reporting would otherwise miss has not been demonstrated; a blind-detection protocol on artefacts with independently characterised defects is outlined in Sec. 5.6 as the decisive next test of practical value.

Planned versus exploratory comparisons. The synthetic-defect responses (Supplementary Table S2, Fig. 5) and the Flick flat detection (Fig. 4) were planned comparisons motivated by the study’s objective of assessing gradient-based descriptor sensitivity. The correlation matrices, power spectral density comparison, Cohen’s d effect sizes, L^p norm convergence analysis, and the cross-descriptor stability comparison (Supplementary Figs. S13–S17) are exploratory and are intended to characterise the descriptor space rather than to test specific hypotheses. This distinction should be borne in mind when interpreting the reported associations. The present study is a proof-of-concept demonstration; generalisability to other artefact types, materials, and measurement instruments remains to be established.

5.4. Limitations

Several limitations should be acknowledged. First, the uncertainty analysis is restricted to within-run Type A repeatability from five sweeps in a single continuous rotation—, complemented by the indicative type-B budget of Sec. 3.8. The small $n = 5$ limits statistical power (relative differences below $\sim 30\%$ cannot be resolved for the FN 111), and between-run reproducibility (independent setups, thermal cycles, physical re-centring of the artefact) has not been assessed.

681 ~~A full GUM-compliant uncertainty budget, incorporating type-B evaluations~~
682 ~~of laser wavelength stability, refractive-index variations, Abbe error, camera~~
683 ~~nonlinearity, and form-removal uncertainty, is beyond the scope of this study;~~
684 the reported expanded uncertainties should be interpreted as lower bounds,
685 and the form-removal systematic (bounded at ≤ 41 nm) is carried outside the
686 quadrature combination.

687 Second, the validation is restricted to two artefacts from a single manufac-
688 turer (JENOPTIK), both of ~ 30 mm diameter;. The two artefacts do differ
689 substantially in material (Al_2O_3 ceramic versus steel) and in residual amplitude
690 (two orders of magnitude in R_a), which provides some breadth, but generalis-
691 ability to other sizes, materials, or diameters, materials such as optical glass
692 or tungsten carbide, and other surface finishes has not been tested. Third,
693 the synthetic-defect analysis uses idealised mathematical defects (isolated spike,
694 uniform scratch, pure sinusoid) injected into the FN 111 ceramic profile only;
695 the response to real manufacturing defects with complex morphologies may dif-
696 fer, and the ~~defect-to-background ratio for~~ scaling analysis of Sec. 5.3 shows
697 quantitatively that full-profile detection sensitivity degrades in proportion to
698 background gradient content (the FN 101 is approximately ratio is ~ 500 times
699 smaller, so the detection sensitivity reported here may not generalise to artefacts
700 with substantially larger form deviations~~less favourable~~). Fourth, BV density
701 and the $W^{1,2}$ seminorm carry intrinsic sampling-interval dependencerequiring
702 ~~conversion to angular units for cross-instrument comparison;~~ the angular-unit
703 convention of Sec. 3.6 standardises reporting, but no inter-laboratory evidence
704 yet exists that the convention transfers across instruments. Fifth, the Flick flat
705 result (Supplementary Fig. S18) is demonstrated on a single artefact with a
706 known, certified macroscopic feature; ~~performance~~ blind-detection performance
707 and performance for subtler, micrometre-scale defects ~~has~~ have not been char-
708 acterised.

709 5.5. Metrological Implications

710 The results carry several implications for profile and roundness metrology
711 practice. The finding that BV density responds to localised gradients that am-
712 plitude averaging may not capture suggests — subject to further validation —
713 a potential role for gradient-based descriptors in applications where isolated
714 defects are of concern; possible examples include quality control of bearing sur-
715 faces, optical components, or precision seals. Because these descriptors are
716 computed from the same residual profile used for ISO ~~4287~~21920-2 parameters,
717 their adoption does not require changes to existing measurement acquisition
718 protocols.

719 The ISO ~~4287 framework~~21920-2 framework (with ISO 4287 as its predecessor)
720 remains the appropriate baseline for surface texture specification and confor-
721 mity assessment. The gradient-based descriptors discussed here are proposed
722 as optional supplementary indicators, not as replacements for standardised pa-
723 rameters. Their metrological utility would be strengthened by inter-laboratory
724 comparisons, by evaluation on a wider range of artefact types, and by ~~the~~
725 ~~development of agreed procedures for converting sampling-dependent units to~~
726 ~~instrument-independent angular gradient units.~~adoption of the angular-unit reporting
727 convention of Sec. 3.6. Until such steps are taken, BV density and related de-
728 scriptors should be regarded as research-grade tools whose diagnostic value has
729 been observed under controlled laboratory conditions on two specific artefacts.

730 5.6. Future Work

731 The present study suggests several directions for further investigation. (i) A
732 between-run reproducibility study with independent setups~~and,~~distinct thermal
733 cycles, and physical re-centring of the artefact would establish whether the
734 gradient-based descriptors retain their relative stability under realistic labo-
735 ratory conditions~~,~~ and would supply the input distributions needed to upgrade
736 the indicative type-B budget of Sec. 3.8 to a complete GUM-compliant evaluation.
737 (ii) Extending the validation to artefacts with different diameters, materials
738 (e.g., ~~glass,~~optical glass, tungsten carbide), and surface finishes would test the

739 generalisability of the findings beyond the two ~30 mm JENOPTIK artefacts
 740 studied here. (iii) A systematic characterisation of BV density response to real
 741 manufacturing defects (scratches, pits, inclusions) of ~~controlled dimensions~~ calibrated
 742 dimensions, culminating in a blind-detection protocol in which the analyst does
 743 not know defect presence or position, would bridge the gap between the present
 744 synthetic-defect analysis and industrial inspection ~~scenarios~~ and would provide
 745 the decisive test of practical value. (iv) An inter-laboratory comparison using
 746 a common artefact set and a standardised processing pipeline would provide
 747 the reproducibility data needed to assess whether BV density could eventually
 748 be considered for standardisation. (v) ~~The development of agreed Empirical~~
 749 verification of the angular-unit ~~conventions for sampling-dependent gradient~~
 750 ~~descriptors would facilitate cross-instrument comparison and potential inclusion~~
 751 convention of Sec. 3.6 across instruments with different angular resolutions
 752 would support the potential inclusion of gradient descriptors in reference soft-
 753 ware for roundness evaluation. (vi) ~~A systematic study of the speed-accuracy~~
 754 trade-off (rotation speed, frame rate, fringe contrast) would establish the shortest
 755 acquisition compatible with the reported uncertainty, a prerequisite for industrial
 756 adoption.

757 Data Availability

758 The analysis code, processed results, and all figures are available at [https:](https://github.com/dawidkucharski/banach-profile-metrology)
 759 [//github.com/dawidkucharski/banach-profile-metrology](https://github.com/dawidkucharski/banach-profile-metrology) (release v1.1.0).
 760 The raw interferometric PNG archives (~32 GB) are available from the corre-
 761 sponding author upon reasonable request.

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